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জন্মশতবার্ষিকী উদ্‌যাপন সফল হোক”



অতিরিক্ত সংখ্যা
কর্তৃপক্ষ কর্তৃক প্রকাশিত

বৃহস্পতিবার ফেব্রুয়ারি ১১, ২০২১

Government of the People's Republic of Bangladesh
Ministry of Housing and Public Works

Notification

Date : 05-11-1426/18-02-2020

S.R.O. No.55-Law/2020.—In exercise of the powers conferred under section 18A of the Building Construction Act, 1952 (Act No. II of 1953) the Government is pleased to make the following Code by repealing the Bangladesh National Building Code, 2006, namely :—

PART I

Chapter 1

Title, Purpose, Scope, Etc

1. **Title and commencement.**—(1) This Code may be called the Bangladesh National Building Code (BNBC) 2020.

(2) It shall come into force at once.

2. **Purpose.**—(1) The purpose of this Code is to establish minimum standards for design, construction, quality of materials, use and occupancy, location and maintenance of all buildings within Bangladesh in order to safeguard, within achievable limits, life, limb, health, property and public welfare.

(2) The installation and use of certain equipment, services and appurtenances related, connected or attached to such buildings are also regulated herein to achieve the same purpose.

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Table 6.1.1: Occupancy Category of Buildings and other Structures for Flood, Surge, Wind and Earthquake Loads.

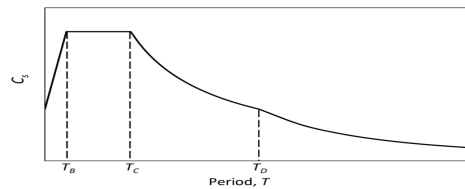
Nature of Occupancy	Occupancy Category
Buildings and other structures that represent a low hazard to human life in the event of failure, including, but not limited to: • Agricultural facilities • Certain temporary facilities • Minor storage facilities	I
All buildings and other structures except those listed in Occupancy Categories I, III and IV	II
Buildings and other structures that represent a substantial hazard to human life in the event of failure, including, but not limited to: • Buildings and other structures where more than 300 people congregate in one area • Buildings and other structures with day care facilities with a capacity greater than 150 • Buildings and other structures with elementary school or secondary school facilities with a capacity greater than 250 • Buildings and other structures with a capacity greater than 500 for colleges or adult education facilities • Healthcare facilities with a capacity of 50 or more resident patients, but not having surgery or emergency Treatment facilities • Jails and detention facilities	III
Buildings and other structures, not included in Occupancy Category IV, with potential to cause a substantial economic impact and/or mass disruption of day-to-day civilian life in the event of failure, including, but not limited to: • Power generating stations^a • Water treatment facilities • Sewage treatment facilities • Telecommunication centers	
Buildings and other structures not included in Occupancy Category IV (including, but not limited to, facilities that manufacture, process, handle, store, use, or dispose of such substances as hazardous fuels, hazardous chemicals, hazardous waste, or explosives) containing sufficient quantities of toxic or explosive substances to be dangerous to the public if released.	

Table 6.2.14: Description of Seismic Zones

Seismic Zone	Location	Seismic Intensity	Seismic Zone Coefficient, Z
1	Southwestern part including Barisal, Khulna, Jessore, Rajshahi	Low	0.12
2	Lower Central and Northwestern part including Noakhali, Dhaka, Pabna, Dinajpur, as well as Southwestern corner including Sundarbans	Moderate	0.20
3	Upper Central and Northwestern part including Brahmanbaria, Sirajganj, Rangpur	Severe	0.28
4	Northeastern part including Sylhet, Mymensingh, Kurigram	Very Severe	0.36

Table 6.2.15: Seismic Zone Coefficient Z for Some Important Towns of Bangladesh

Town	Z	Town	Z	Town	Z	Town	Z
Bagerhat	0.12	Gaibandha	0.28	Magura	0.12	Patuakhali	0.12
Bandarban	0.28	Gazipur	0.20	Manikganj	0.20	Pirojpur	0.12
Barguna	0.12	Gopalganj	0.12	Maulvibazar	0.36	Rajbari	0.20
Barisal	0.12	Habiganj	0.36	Meherpur	0.12	Rajshahi	0.12
Bhola	0.12	Jaipurhat	0.20	Mongla	0.12	Rangamati	0.28
Bogra	0.28	Jamalpur	0.36	Munshiganj	0.20	Rangpur	0.28
Brahmanbaria	0.28	Jessore	0.12	Mymensingh	0.36	Satkhira	0.12
Chandpur	0.20	Jhalokati	0.12	Narail	0.12	Shariatpur	0.20
Chapainababganj	0.12	Jhenaidah	0.12	Narayanganj	0.20	Sherpur	0.36
Chittagong	0.28	Khagrachari	0.28	Narsingdi	0.28	Sirajganj	0.28
Chuadanga	0.12	Khulna	0.12	Natore	0.20	Srimangal	0.36
Comilla	0.20	Kishoreganj	0.36	Naogaon	0.20	Sunamganj	0.36
Cox's Bazar	0.28	Kurigram	0.36	Netrakona	0.36	Sylhet	0.36
Dhaka	0.20	Kushtia	0.20	Nilphamari	0.12	Tangail	0.28
Dinajpur	0.20	Lakshmipur	0.20	Noakhali	0.20	Thakurgaon	0.20
Faridpur	0.20	Lalmanirhat	0.28	Pabna	0.20		
Feni	0.20	Madaripur	0.20	Panchagarh	0.20		

Figure 6.2.25 Typical shape of the elastic response spectrum coefficient C_s

2.5.5.2 Seismic design category

Buildings shall be assigned a seismic design category among B, C or D based on seismic zone, local site conditions and importance class of building, as given in Table 6.2.18. Seismic design category D has the most stringent seismic design detailing, while seismic design category B has the least seismic design detailing requirements.

Table 6.2.18: Seismic Design Category of Buildings

Site Class	Occupancy Category I, II and III				Occupancy Category IV			
	Zone 1	Zone 2	Zone 3	Zone 4	Zone 1	Zone 2	Zone 3	Zone 4
SA	B	C	C	D	C	D	D	D
SB	B	C	D	D	C	D	D	D
SC	B	C	D	D	C	D	D	D
SD	C	D	D	D	D	D	D	D
SE, S ₁ , S ₂	D	D	D	D	D	D	D	D

2.5.5.3 Building irregularity

Buildings with irregularity in plan or elevation suffer much more damage in earthquakes than buildings with regular configuration. A building may be considered as irregular, if at least one of the conditions given below are applicable:

2.5.5.3.1 Plan irregularity: Following are the different types of irregularities that may exist in the plan of a building.

(i) Torsion irregularity

To be considered for rigid floor diaphragms, when the maximum storey drift (Δ_{max}) as shown in Figure 6.2.27(a), computed including accidental torsion, at one end of the structure is more than 1.2 times the average ($\Delta_{avg} = \frac{\Delta_{max} + \Delta_{min}}{2}$) of the storey drifts at the two ends of the structure. If $\Delta_{max} > 1.4\Delta_{avg}$ then the irregularity is termed as extreme torsional irregularity.

(ii) Re-entrant corners

Both projections of the structure beyond a re-entrant corner [Figure 6.2.27(b)] are greater than 15 percent of its plan dimension in the given direction.

6.3.9 Limits of Reinforcement for Compression Members

6.3.9.1 For noncomposite compression members, the area of longitudinal reinforcement, A_{st} , shall be not less than $0.01A_g$ or more than $0.06A_g$. To avoid practical difficulties in placing and compacting of concrete as well as to deliver ductility to noncomposite compression members, area of longitudinal reinforcement, A_{st} , is preferred not to exceed $0.04A_g$ unless absolutely essential.

6.3.9.2 Minimum number of longitudinal bars in compression members shall be 4 for bars within rectangular or circular ties, 3 for bars within triangular ties, and 6 for bars enclosed by spirals conforming to Sec 6.3.9.3.

6.3.9.3 Volumetric spiral reinforcement ratio, ρ_s , shall be not less than the value given by

$$\rho_s = 0.45 \left(\frac{A_g}{A_{ch}} - 1 \right) \frac{f'_c}{f_{yt}} \quad (6.6.12)$$

Where the value of f_{yt} used in Eq. 6.6.12 shall not exceed 700 MPa. For f_{yt} greater than 420 MPa, lap splices according to 8.1.9.3(e) shall not be used.

6.3.10 Slenderness Effects in Compression Members

6.3.10.1 Slenderness effects shall be permitted to be neglected in the following cases:

- (a) for compression members not braced against side sway when:

$$\frac{kl_u}{r} \leq 22 \quad (6.6.13)$$

- (b) for compression members braced against side sway when:

$$\frac{kl_u}{r} \leq 34 - 12(M_1/M_2) \leq 40 \quad (6.6.14)$$

Where, M_1/M_2 is positive if the column is bent in single curvature, and negative if the member is bent in double curvature.

Compression members may be considered to be braced against side sway when bracing elements have a total stiffness, resisting lateral movement of that story, of at least 12 times the gross stiffness of the columns within the story.

The Jackson and Moreland Alignment Charts (Figure 6.6.5), which allow a graphical determination of k for a column of constant cross section in a multibay frame may be used as the primary design aid to estimate the effective length factor k .

settlement, creep, shrinkage, and temperature change, based on a realistic assessment of such effects occurring in service.

6.4.3.5.3 Where shear reinforcement is required by Sec 6.4.3.5.1 or for strength and where Sec 6.4.4.1 allows torsion to be neglected, $A_{v,min}$ shall be computed by

$$A_{v,min} = 0.062 \sqrt{f'_c} \frac{b_w s}{f_{yt}} \quad (6.6.55)$$

But, shall not be less than $(0.35b_w s)/f_{yt}$.

6.4.3.6 Design of shear reinforcement

6.4.3.6.1 Where V_u exceeds ϕV_c , shear reinforcement shall be provided to satisfy Equations 6.6.47 and 6.6.48, where V_s shall be computed in accordance with Sections 6.4.3.6.2 to 6.4.3.6.9.

6.4.3.6.2 Where shear reinforcement perpendicular to axis of member is used,

$$V_s = \frac{A_v f_{yt} d}{s} \quad (6.6.56)$$

Where, A_v is the area of shear reinforcement within spacing s .

6.4.3.6.3 Where circular ties, hoops, or spirals are used as shear reinforcement, V_s shall be computed using Eq. 6.6.56 where d is defined in Sec 6.4.2.3 for circular members, A_v shall be taken as two times the area of the bar in a circular tie, hoop, or spiral at a spacing S , s is measured in a direction parallel to longitudinal reinforcement, and f_{yt} is the specified yield strength of circular tie, hoop, or spiral reinforcement.

6.4.3.6.4 Where inclined stirrups are used as shear reinforcement,

$$V_s = \frac{A_v f_{yt} (\sin \alpha + \cos \alpha) d}{s} \quad (6.6.57)$$

Where, α is angle between inclined stirrups and longitudinal axis of the member, and s is measured in direction parallel to longitudinal reinforcement.

6.4.3.6.5 Where shear reinforcement consists of a single bar or a single group of parallel bars, all bent up at the same distance from the support,

$$V_s = A_v f_y \sin \alpha \quad (6.6.58)$$

But, not greater than $0.25\sqrt{f'_c}b_w d$, where α is angle between bent-up reinforcement and longitudinal axis of the member.

6.4.8.9.4 Ratio of vertical shear reinforcement area to gross concrete area of horizontal section, ρ_t shall not be less than the larger of 0.0025 and the value obtained from:

$$\rho_t = 0.0025 + 0.5 \left(2.5 - \frac{h_w}{l_w} \right) (\rho_t - 0.0025) \quad (6.6.71)$$

The value of ρ_t calculated by Eq. 6.6.71 need not be greater than ρ_t required by Sec 6.4.8.9.1. In Eq. 6.6.71, l_w is the overall length of the wall, and h_w is the overall height of the wall.

6.4.8.9.5 Spacing of vertical shear reinforcement shall not exceed the smallest of $\frac{l_w}{3}$, $3h$, and 450 mm, where l_w is the overall length of the wall.

6.4.9 Transfer of Moments to Columns

6.4.9.1 When gravity load, wind, earthquake, or other lateral forces cause transfer of moment at connections of framing elements to columns, the shear resulting from moment transfer shall be considered in the design of lateral reinforcement in the columns.

6.4.9.2 Except for connections not part of a primary seismic load-resisting system that are restrained on four sides by beams or slabs of approximately equal depth, connections shall have lateral reinforcement not less than that required by Eq. 6.6.55 within the column for a depth not less than that of the deepest connection of framing elements to the columns. See also Sec. 8.1.13 Chapter 8.

6.4.10 Provisions for Footings and Slabs

6.4.10.1 The shear strength of footings and slabs in the vicinity of columns, concentrated loads, or reactions is governed by the more severe of the following two conditions:

6.4.10.1.1 Beam action where each critical section to be investigated extends in a plane across the entire width. The slab or footing shall be designed in accordance with Sections 6.4.1 to 6.4.3 for beam action.

6.4.10.1.2 For two-way action, each of the critical sections to be investigated shall be located so that its perimeter b_o is a minimum but need not approach closer than $\frac{d}{2}$ to:

- (a) Edges or corners of columns, concentrated loads, or reaction areas; and
- (b) Changes in slab thickness such as edges of capitals, drop panels, or shear caps.

For two-way action, the slab or footing shall be designed in accordance with Sections 6.4.10.2 to 6.4.10.6.

6.4.10.1.3 For square or rectangular columns, concentrated loads, or reaction areas, the critical sections with four straight sides shall be permitted.

PART VI
Chapter 8

Detailing of Reinforcement in Concrete Structures

8.1 Introduction

Provisions of Sections 8.1 and 8.2 of Chapter 8 shall apply for detailing of reinforcement in reinforced concrete members, in general. For reinforced concrete structures, subject to earthquake loadings in seismic design categories B, C and D, special provisions contained in Sec 8.3 of this Chapter shall apply. The definitions and notation provided in the following Sections are related to Sec 8.3. The definitions and notation used in other Sections, unless otherwise mentioned, are similar to those provided in Sections 6.1.1 and 6.1.2 Chapter 6.

8.1.1 Definitions and Notation

8.1.1.1 Definitions

BASE OF STRUCTURE	The level at which earthquake motions are assumed to be imparted to a structure. This level does not necessarily coincide with the ground level.
BOUNDARY MEMBERS	Members along wall and diaphragm edges strengthened by longitudinal and transverse reinforcement. These members do not necessarily require an increase in the thickness of the wall or diaphragm. If required, edges of openings within walls and diaphragms shall be provided with boundary members.
COLLECTOR ELEMENTS	Elements that are used to transmit the inertial forces within the diaphragms to members of the lateral force resisting systems.
CROSS TIE	A continuous bar having a hook not less than 135° with at least a six diameter extension at one end but not less than 75 mm, and a hook not less than 90° with at least a six diameter extension at the other end. The hooks shall engage peripheral longitudinal bars. The 90° hooks of two successive cross ties engaging the same longitudinal bars shall be alternated end for end.

DEVELOPMENT LENGTH OF A STANDARD HOOK	The shortest distance between the critical section and a tangent to the outer edge of the 90° hook.
HOOP	A hoop is a closed tie or continuously round tie. A closed tie can be made up of several reinforcing elements with 135° hooks having a six diameter extension at each end (but not less than 75 mm). A continuously round tie shall have at each end a 135° hook with a six diameter extension that engages the longitudinal reinforcement but not less than 75 mm.
LATERAL FORCE RESISTING SYSTEM	That portion of the structure composed of members designed to resist forces related to earthquake effects.
SHELL CONCRETE	Concrete outside the transverse reinforcement confining the concrete
STRUCTURAL DIAPHRAGMS	Structural members, such as floor and roof slabs, which transmit inertial forces to lateral force resisting members.
STRUCTURAL WALLS	Walls designed to resist combinations of shears, moments, and axial forces induced by earthquake motions. A shear wall is a structural wall.
STRUT	An element of a structural diaphragm used to provide continuity around an opening in the diaphragm.
TIE ELEMENTS	Elements used to transmit inertial forces and prevent separation of building components.

8.1.1.2 Notation

A_{ch}	=	Cross-sectional area of a structural member measured out to out of transverse reinforcement, mm ²
A_{cp}	=	Area of concrete section resisting shear of an individual pier or horizontal wall segment, mm ²
A_{cv}	=	Net area of concrete section bounded by web thickness and length of section in the direction of shear force considered, mm ²
A_g	=	Gross area of section, mm ²
A_j	=	Effective cross-sectional area within a joint, see Sec 8.3.7.3, in a plane parallel to plane of reinforcement generating shear in the joint. The joint depth shall be the overall depth of the column. Where a beam frames into a support of larger width, the effective width of the joint shall not exceed the smaller of : (a) Beam width plus the joint depth (b) twice the smaller perpendicular distance from the longitudinal axis of the beam to the column side (See Sec 8.3.7.3)

A_{sh}	=	Total cross-sectional area of transverse reinforcement (including cross ties) within spacing s and perpendicular to dimension h_c
E	=	Load effects of earthquake or related internal moments and forces
M_{pr}	=	Probable flexural moment strength of members, with or without axial load, determined using the properties of the member at the joint faces assuming a tensile strength in the longitudinal bars of at least $1.25f_y$ and a strength reduction factor ϕ of 1.0, N-mm
M_s	=	Portion of slab moment balanced by support moment
V_c	=	Nominal shear strength provided by concrete, N
V_e	=	Design shear force corresponding to the development of the probable moment strength of the member, N
V_n	=	Nominal shear strength, N
V_u	=	Factored shear force at section, N
b	=	Effective compressive flange width of a structural member, mm
b_w	=	Web width or diameter of circular section, mm
d	=	Distance from extreme compression fibre to centroid of longitudinal tension reinforcement, mm
d_b	=	Bar diameter, mm
f'_c	=	Specified compressive strength of concrete, MPa
f_y	=	Specified yield strength of reinforcement, MPa
f_{yt}	=	Specified yield strength of transverse reinforcement, MPa
h	=	Overall thickness or height of member, mm
h_c	=	Cross-sectional dimension of column core measured to the outside edge of the transverse reinforcement composing area A_{sh} mm centre to centre of confining reinforcement
h_w	=	Height of entire wall (diaphragm) or of the segment of wall (diaphragm) considered, mm
h_x	=	Maximum centre to centre horizontal spacing of crossties or hoop legs on all faces of the column, mm
l_d	=	Development length in tension of deformed bar, deformed wire, plain and deformed welded wire reinforcement, mm

- l_{dh} = Development length in tension of deformed bar or deformed wire with a standard hook, measured from critical section to outside end of hook [straight embedment length between critical section and start of hook (point of tangent) plus inside radius of bend and one bar diameter], mm
- l_o = Minimum length, measured from joint face along axis of structural member, over which special transverse reinforcement must be provided, mm
- l_w = Length of entire wall (diaphragm) or of segment of wall (diaphragm) considered in the direction of shear force, mm
- s = Spacing of transverse reinforcement measured along longitudinal axis of the structural member, mm
- s_o = Maximum spacing of transverse reinforcement, mm
- α_c = Coefficient defining the relative contribution of concrete strength to wall strength
- ρ = Ratio of tension reinforcement to member area = A_s/b_d
- ρ_g = Ratio of total reinforcement area to cross-sectional area of column
- ρ_n = Ratio of distributed shear reinforcement on a plane perpendicular to plane of A_{cv}
- ρ_s = Ratio of volume of spiral reinforcement to the core volume confined by the spiral reinforcement (measured out to out of spiral)
- ρ_v = A_{sv}/A_{cv} ; where A_{sv} is the projection on A_{cv} of area of distributed shear reinforcement crossing the plane of A_{cv}
- ϕ = Strength reduction factor.

8.1.2 Standard Hooks and Minimum Bend Diameters

8.1.2.1 Standard hooks

The term "standard hook" as used in this Code shall mean one of the following:

- 180° bend plus an extension of at least 4 bar diameters, but not less than 65 mm at the free end of the bar.
- 90° bend plus an extension of at least 12 bar diameters at the free end of the bar.

- (c) For stirrup and tie anchorage
- (i) For 16 mm diameter bar and smaller, a 90° bend plus an extension of at least 6 bar diameters at the free end of the bar,
 - (ii) For 19 mm to 25 mm diameter bars, a 90° bend plus an extension of at least 12 bar diameters at the free end of the bar,
 - (iii) For 25 mm diameter bar and smaller, a 135° bend plus an extension of at least 6 bar diameters at the free end of the bar,
 - (iv) For closed ties and continuously wound ties, a 135° bend plus an extension of at least 6 bar diameters, but not less than 75 mm.
- (d) Seismic hook is defined as a hook on a stirrup, hoop, or crosstie having a bend not less than 135°, except that circular hoops shall have a bend not less than 90°. Hooks shall have a six-diameter (but not less than 75 mm) extension that engages the longitudinal reinforcement and projects into the interior of the stirrup or hoop.

8.1.2.2 Minimum bend diameters

- (a) The minimum diameter of bend measured on the inside of the bar, for standard hooks other than for stirrups and ties in sizes of 10 mm to 16 mm diameter shall not be less than the values shown in Table 6.8.1.

Table 6.8.1: Minimum Diameters of Bend

Bar Size	Minimum Diameter of Bend
10 mm $\leq d_b \leq$ 25 mm	$6d_b$
25 mm $< d_b \leq$ 40 mm	$8d_b$
40 mm $< d_b \leq$ 57 mm	$10d_b$

- (b) For stirrups and tie hooks, inside diameter of bend shall not be less than 4 bar diameters for 16 mm diameter bar and smaller. For bars larger than 16 mm diameter, bend diameter shall be in accordance with Table 6.8.1.
- (c) Inside diameter of bend in welded wire reinforcement for stirrups and ties shall not be less than 4 bar diameters for deformed wire larger than ASTM MD40 size (ASTM A1022) and 2 bar diameters for all other wires. Bends with inside diameter of less than 8 bar diameters shall not be less than 4 bar diameters from nearest welded intersection.

8.1.9 Lateral Reinforcement for Columns

8.1.9.1 Lateral reinforcement for compression members shall conform to the provisions of Sections 8.1.9.3 and 8.1.9.4 below and where shear or torsion reinforcement is required, shall also conform to provisions of Sec 6.4.

8.1.9.2 Lateral reinforcement requirements for composite columns shall conform to Sections 6.3.13.7 and 6.3.13.8 Chapter 6.

8.1.9.3 Spirals: Spiral reinforcement for columns shall conform to Sec 6.3.9.3 Chapter 6 and to the following:

- (a) Spirals shall consist of evenly spaced continuous bar or wire of such size and so assembled as to permit handling and placing without distortion from designed dimensions.
- (b) Size of spirals shall not be less than 10 mm diameter for cast-in-place construction.
- (c) The minimum and maximum clear spacing between spirals shall be 25 mm and 75 mm respectively.
- (d) Anchorage of spiral reinforcement shall be provided by 1.5 extra turns of spiral bar or wire at each end of a spiral unit.
- (e) Splices in spiral reinforcement shall be lap splices of 48 spiral diameter for deformed uncoated bar or wire and 72 spiral diameter for other cases, but not less than 300 mm.
- (f) Spirals shall extend from the top of footing or slab in any storey to the level of the lowest horizontal reinforcement in members supported above.
- (g) Spirals shall extend above termination of spiral to bottom of slab or drop panel, where beams or brackets do not frame into all sides of a column.
- (h) Spirals shall extend to a level at which the diameter or width of capital is 2 times that of the column, in case of columns with capitals.
- (i) Spirals shall be held firmly in place and true to line.

8.1.9.4 Ties: Tie reinforcement for compression members shall conform to the following:

- (a) All bars shall be enclosed by lateral ties, at least 10 mm diameter in size for longitudinal bars 32 mm diameter or smaller, and at least 12 mm diameter in size for 36 mm to 57 mm diameter and bundled longitudinal bars.
- (b) Vertical spacing of ties shall not exceed 16 longitudinal bar diameters or 48 tie diameters, or the least dimension of the compression members.

- (c) Ties shall be arranged such that every corner and alternate longitudinal bar shall have lateral support provided by the corner of a tie with an included angle not more than 135° . No vertical bar shall be farther than 150 mm clear on each side along the tie from such a laterally supported bar. Where longitudinal bars are located around the perimeter of a circle, a complete circular tie is allowed.
- (d) The lowest tie in any storey shall be placed within one-half the required tie spacing from the top most horizontal reinforcement in the slab or footing below. The uppermost tie in any storey shall be within one-half the required tie spacing from the lowest horizontal reinforcement in the slab or drop panel above.
- (e) Where beams or brackets provide concrete confinement at the top of the column on all (four) sides, top tie shall be within 75 mm of the lowest horizontal reinforcement in the shallowest of such beams or brackets.
- (f) Where anchor bolts are placed in the top of columns or pedestals, the bolts shall be enclosed by lateral reinforcement that also surrounds at least four vertical bars of the column or pedestal. The lateral reinforcement shall be distributed within 125 mm of the top of the column or pedestal, and shall consist of at least two 12 mm diameter bars or three 10 mm diameter bars.
- (g) Where longitudinal bars are arranged in a circular pattern, individual circular ties per specified spacing may be used.

8.1.10 Lateral Reinforcement for Beams

8.1.10.1 Compression reinforcement in beams shall be enclosed by ties or stirrups satisfying the size and spacing limitations in Sec 8.1.9.4 above. Such ties or stirrups shall be provided throughout the distance where compression reinforcement is required.

8.1.10.2 Lateral reinforcement for flexural framing members subject to stress reversals or to torsion at supports shall consist of closed ties, closed stirrups, or spirals extending around the flexural reinforcement.

8.1.10.3 Closed ties or stirrups shall be formed in one piece by overlapping standard stirrup or tie end hooks around a longitudinal bar, or formed in one or two pieces lap spliced with a Class B splice (lap of $1.3l_d$) or anchored in accordance with Sec 8.2.10.

8.1.11 Shrinkage and Temperature Reinforcement

8.1.11.1 Where the flexural reinforcement extends in one direction only, reinforcement for shrinkage and temperature stresses shall be provided perpendicular to flexural reinforcement in structural slabs. Shrinkage and temperature reinforcement shall be provided in accordance with Sec 8.1.11.2 below.

8.2.20.2 Lap splices of welded deformed wire reinforcement, with no cross wires within the lap splice length, shall be determined as for deformed wire.

8.2.20.3 Where any plain wires, or deformed wires larger than ASTM MD200, are present in the welded deformed wire reinforcement in the direction of the lap splice or where welded deformed wire reinforcement is lap spliced to welded plain wire reinforcement, reinforcement shall be lap spliced in accordance with Sec 8.2.21.

8.2.21 Splices of Welded Plain Wire Reinforcement in Tension

Minimum length of lap for lap splices of welded plain wire reinforcement shall be in accordance with Sections 8.2.21.1 and 8.2.21.2.

8.2.21.1 Where A_s provided is less than twice that required by analysis at splice location, length of overlap measured between outermost cross wires of each reinforcement sheet shall be not less than the largest of one spacing of cross wires plus 50 mm, $1.5l_d$ and 150 mm, where l_d is calculated in accordance with Sec 8.2.19 to develop f_y .

8.2.21.2 Where A_s provided is at least twice that required by analysis at splice location, length of overlap measured between outermost cross wires of each reinforcement sheet shall not be less than the larger of $1.5l_d$ and 50 mm, where l_d is calculated in accordance with Sec 8.2.19 to develop f_y .

8.3 Earthquake-Resistant Design Provisions

8.3.1 Scope

This section contains special requirements for design and construction of reinforced concrete members of a structure for which the design forces, related to earthquake motions, have been determined on the basis of energy dissipation in the nonlinear range of response.

8.3.2 Provisions

- (a) The provisions of Chapter 6, shall apply except as modified by the provisions of this Section.
- (b) Structures assigned to seismic design category SDC D (see Chapter 2), all reinforced concrete structures shall satisfy the requirements of special seismic detailing as given in Sections 8.3.3 to 8.3.8 in addition to the requirements of Chapter 6. The provisions for special moment frames shall not permit the use of slab without beam as part of seismic force-resisting system.

- (c) Structures assigned to SDC C (see Chapter 2), all reinforced concrete structures shall be built to satisfy the requirements of intermediate seismic detailing as given in Sec 8.3.10 in addition to the requirements of Chapter 6.
- (d) Structures assigned to SDC B (see Chapter 2), all reinforced concrete structures shall be built to satisfy the requirements of ordinary detailing as given in Sec 8.3.9 in addition to the requirements of Chapter 6.
- (e) Structures in lower SDCs are permitted to design with detailing provisions of higher SDCs to take advantage of lower design force levels.

8.3.3 General Requirements

8.3.3.1 Analysis and proportioning of structural members

- (a) The interaction of all structural and nonstructural members shall be considered in the analysis.
- (b) Rigid members which are not a part of the lateral force resisting system are allowed provided their effect on the response of the system is considered and accommodated in the structural design. Consequences of failure of structural and nonstructural members which are not a part of the lateral force resisting system shall also be considered.
- (c) Structural members below base of structure required to transmit forces resulting from earthquake effects to the foundation shall also comply with the requirements of this section.
- (d) All structural members which are not a part of the lateral force resisting system shall conform to Sec 8.3.9.

8.3.3.2 Strength reduction factors

Strength reduction factors shall be in accordance with Sections 6.2.3.2 to 6.2.3.4.

8.3.3.3 Concrete in special moment frames and special structural walls

Compressive strength f'_c of the concrete shall be not less than 21 N/mm². Specified compressive strength of light-weight concrete, f'_c shall not exceed 35MPa unless demonstrated by experimental evidence. Modification factor λ for lightweight concrete in Sec 8.3 shall be in accordance with Sec 6.1.8 unless noted otherwise.

8.3.3.4 Reinforcement in special moment frames and special structural walls

- (a) Requirements of Sec 8.3.3.4 shall apply to special moment frames, special structural walls and all components of special structural walls including coupling beams and wall piers.

8.3.5 Special Moment Frame Members Subjected to Bending and Axial Load

8.3.5.1 Scope

The requirements of this section shall apply to columns and other frame members serving to resist earthquake forces and having a factored axial force exceeding $0.1A_g f'_c$. These frame members shall also satisfy the following conditions. The requirements are also shown in Figure 6.8.6.

- (a) The shortest cross-sectional dimension shall not be less than 300 mm.
- (b) The ratio of the shortest cross-sectional dimension to the perpendicular dimension shall not be less than 0.4.

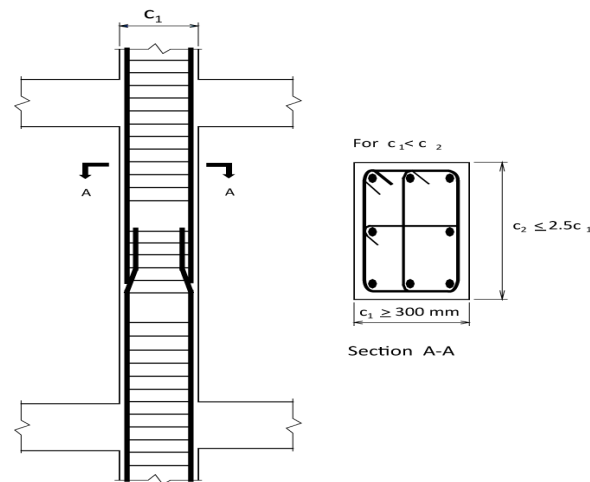


Figure 6.8.6 General requirements for special moment frames subjected to bending and axial load (Sec 8.3.5.1)

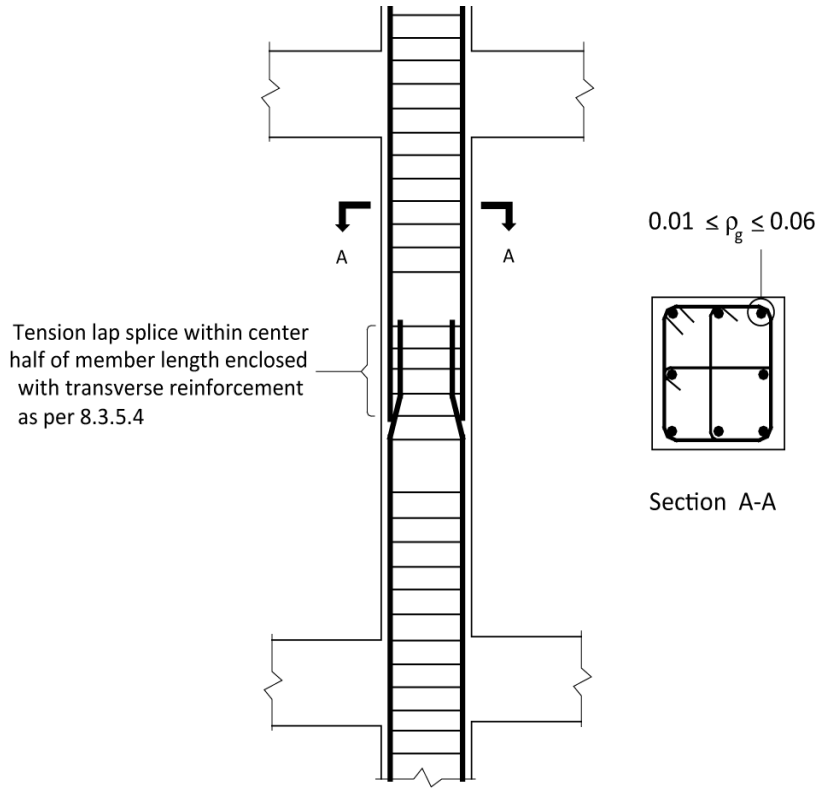


Figure 6.8.7 Longitudinal reinforcement requirements (SMF) (Sec 8.3.5.3)

8.3.5.4 Transverse reinforcement

- (a) Transverse reinforcement shall be provided as specified below and shown in Figures 6.8.8 and 6.8.9 unless a larger amount is required by Sec 8.3.8.
- (i) The volumetric ratio of spiral or circular hoop reinforcement, ρ_s shall not be less than that indicated by the following equation:

$$\rho_s = \frac{0.12f'_c}{f_{yt}} \quad (6.8.6)$$

and shall not be less than that required by Eq. (6.6.12).

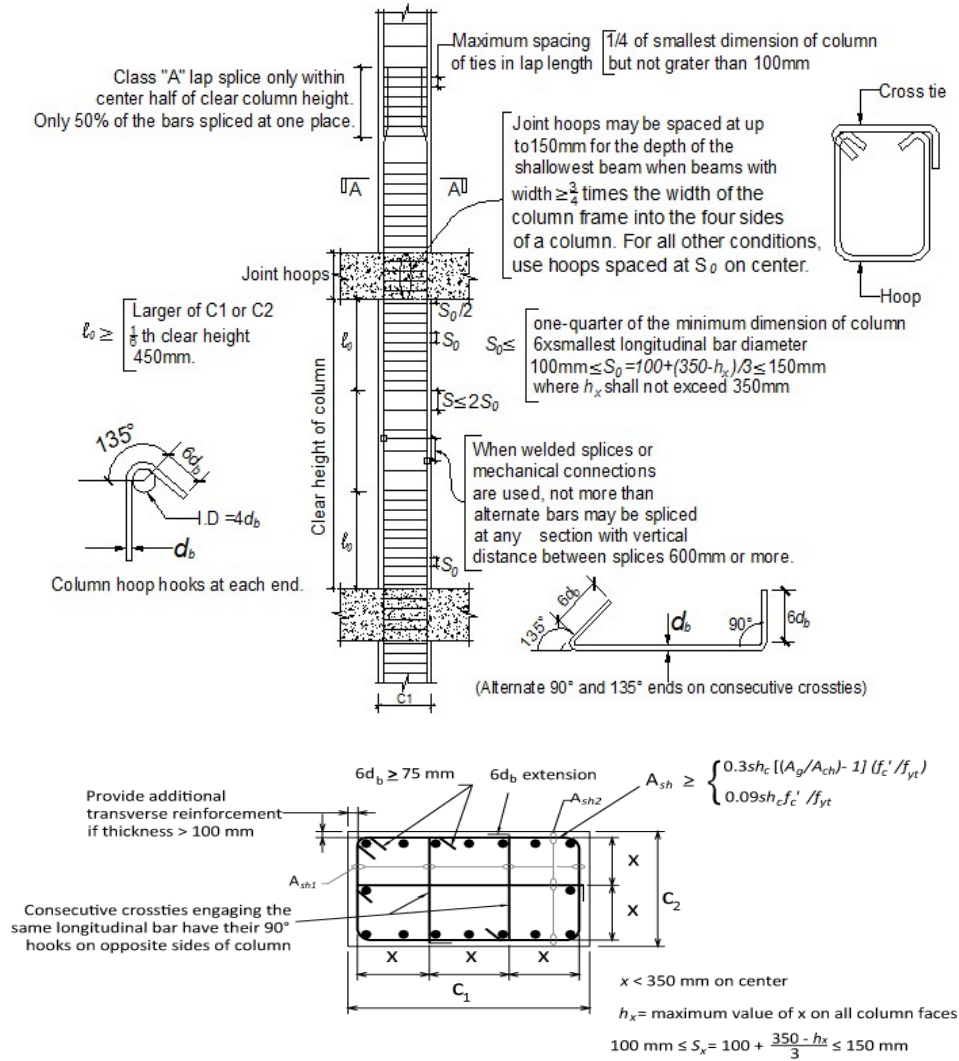
- (ii) The total cross-sectional area of rectangular hoop reinforcement shall not be less than that given by the following equations:

$$A_{sh} = 0.3 \left(\frac{shcf'_c}{f_{yt}} \right) \left(\frac{A_g}{A_{ch}} - 1 \right) \quad (6.8.7)$$

$$A_{sh} = 0.09 \left(\frac{shcf'_c}{f_{yt}} \right) \quad (6.8.8)$$

- (iii) Transverse reinforcement shall be provided by either single or overlapping hoops or cross ties of the same bar size and spacing. Each end of the cross ties shall engage a peripheral longitudinal reinforcing bar. Consecutive cross ties shall be alternated end for end along the longitudinal reinforcement.
- (iv) If the design strength of member core satisfies the requirements of the specified loading combinations including earthquake effect, Eq. 6.8.7 and Eq. 6.6.12 need not be satisfied.
- (b) Spacing of transverse reinforcement along the length l_o of the member shall not exceed the smallest of (i) one-quarter of the minimum dimension (ii) six times the diameter of the smallest longitudinal bar and (iii) $s_o = 100 + \frac{(350-h_x)}{3}$. The value of s_o shall not exceed 150 mm and need not be taken less than 100 mm.
- (c) Spacing of cross ties or legs of overlapping hoops shall not be more than 350 mm on centre in the direction perpendicular to the longitudinal axis of the member.
- (d) The volume of transverse reinforcement in amount specified in (a) through (c) above shall be provided over a length l_o from each joint face and on both sides of any section where flexural yielding is likely to occur in connection with inelastic lateral displacements of the frame. The length l_o shall not be less than (i) the depth of the member at the joint face or at the section where flexural yielding is likely to occur, (ii) one-sixth of the clear span of the member, and (iii) 450 mm.
- (e) If the factored axial force in columns supporting reactions from discontinued stiff members, such as walls, exceeds $0.1A_gf'_c$ they shall be provided with transverse reinforcement as specified in (a) through (c) above over their full height beneath the level at which the discontinuity occurs. Transverse reinforcement shall extend into the discontinued member for at least the development length of the largest longitudinal reinforcement in the column in accordance with Sec 8.3.7.4. If the lower end of the column terminates on a wall, transverse reinforcement as specified above shall extend into the wall for at least the development length of the largest longitudinal reinforcement in the column at the point of termination. If the column terminates on a footing or mat, transverse reinforcement as specified in above shall extend at least 300 mm into the footing or mat.

- (f) Where transverse reinforcement as specified in (a) through (c) above, is not provided throughout the full length of the column, the remainder of the column length shall contain spiral or hoop reinforcement with centre to centre spacing not exceeding the smaller of 6 times the diameter of the longitudinal column bars or 150 mm.



Note: In beam column joints where members frame into all four sides of the joint and each member width is at least three-fourths the column width, the spacing of transverse reinforcement shall be 150 mm within the overall depth of the shallowest frame member. For all other conditions spacing shall be S_0 . Use hoops and cross ties in beam column joint.

Figure 6.8.8 Transverse reinforcement requirements- rectangular hoop for members subjected to bending and axial load rectangular hoop (SMF) (Sec 8.3.5.4)

8.3.7.2 Transverse reinforcement

The provisions of transverse reinforcement are shown in Figures 6.8.15 and 6.8.16, stated as under.

- As specified in Sec 8.3.5.4, transverse hoop reinforcement shall be provided within the joint, unless the joint is confined by structural members as specified in (b) below.
- Within the depth of the shallowest framing member, transverse reinforcement equal to at least one-half the amount required by Sec 8.3.5.4(a) shall be provided where members frame into all four sides of the joint and where each member width is at least three-fourths the column width. At these locations, the spacing specified in Sec 8.3.5.4(b) may be increased to 150 mm.
- As required by Sec 8.3.5.4, transverse reinforcement shall be provided through the joint to provide confinement for longitudinal beam reinforcement outside the column core if such confinement is not provided by a beam framing into the joint.

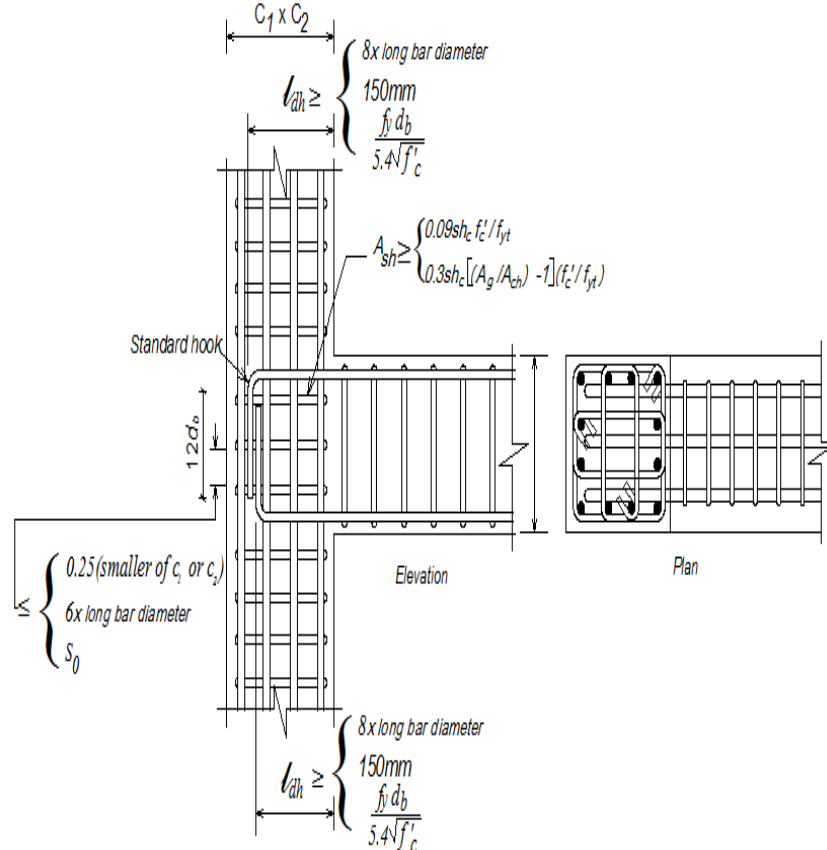


Figure 6.8.15 General requirements and transverse reinforcement requirements for joints not confined by structural member

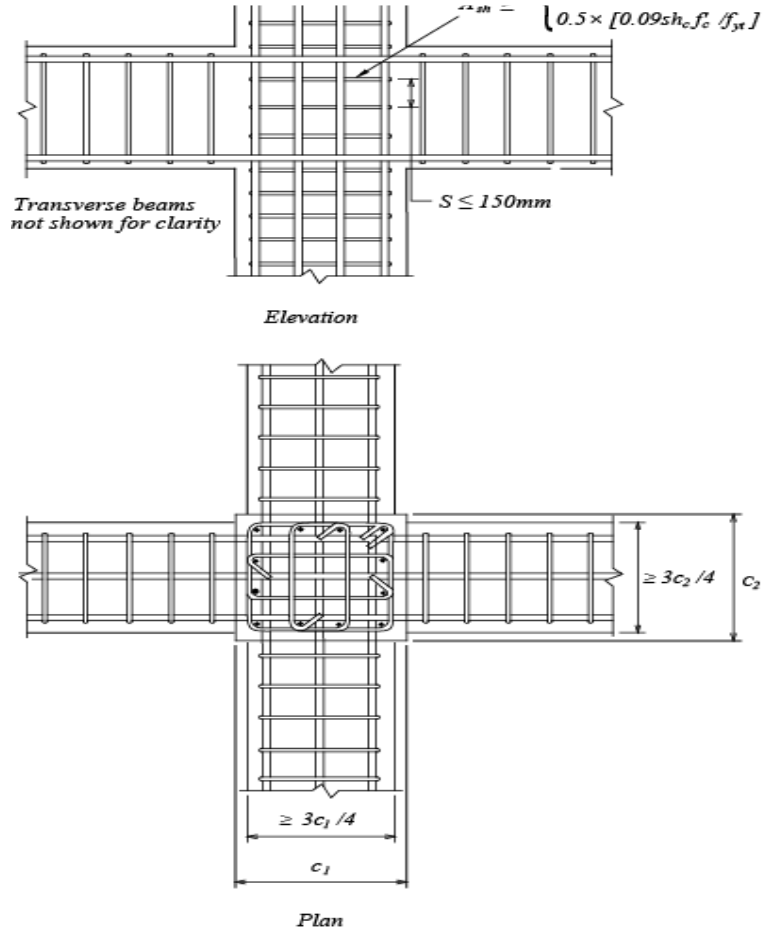


Figure 6.8.16 Transverse reinforcement requirements for joints confined by structural member

8.3.7.3 Shear Strength

The nominal shear strength for the joint shall be taken not greater than the forces specified below:

Joints confined on all four faces: $1.7\sqrt{f'_c} A_j$

Joints confined on three faces or on two opposite faces: $1.2\sqrt{f'_c} A_j$

Others: $1.0\sqrt{f'_c} A_j$

A member that frames into a face is considered to provide confinement to the joint if at least three-quarters of the face of the joint is covered by the framing member. A joint is considered to be confined if such confining members frame into all faces of the joint.

- (f) Nominal shear strength of all wall piers sharing a common lateral force shall not be assumed to exceed $0.67A_{cv}\sqrt{f'_c}$, where A_{cv} is the total cross-sectional area, and the nominal shear strength of any one of the individual wall piers shall not be assumed to exceed $0.83A_{cp}\sqrt{f'_c}$

Where A_{cp} represents the cross-sectional area of the pier considered.

- (g) Nominal shear strength of horizontal wall segments shall be assumed not to exceed $0.83A_{cp}\sqrt{f'_c}$ where A_{cp} represents the cross-sectional area of a horizontal wall segment.

8.3.9 Ordinary Moment Frame Members not Proportioned to Resist Forces Induced by Earthquake Motion

8.3.9.1 Induced moments

Frame members assumed not to contribute to lateral resistance shall be detailed according to (a) or (b) below depending on the magnitude of moments induced in those members when subjected to twice the lateral displacement under the factored lateral forces.

- (a) Members with factored gravity axial forces not exceeding $0.1A_gf'_c$ shall satisfy Sections 8.3.4.2(a) and 8.3.8.1(a) and members with factored gravity axial forces exceeding $0.1A_gf'_c$ shall satisfy Sections 8.3.5.4, 8.3.7.2(a) and 8.3.8.1(b) when the induced moment exceeds the design moment strength of the frame member.
- (b) The member shall satisfy Sec 8.3.4.2(a) when the induced moment does not exceed the design moment strength of the frame members.

8.3.9.2 Tie requirements

All frame members with factored axial compressive forces exceeding $0.1A_gf'_c$ shall satisfy the following special requirements unless they comply with Sec 8.3.5.4.

- (a) Ties shall have hooks not less than 135° with extensions not less than 6 tie bar diameter or 60 mm. Cross ties as defined in Sec 8.3.2 are allowed.
- (b) The maximum tie spacing shall be s_0 over a length l_0 measured from the joint face. The spacing s_0 shall be not more than (i) eight diameters of the smallest longitudinal bar enclosed, (ii) 24 tie bar diameters, and (iii) one-half the least cross-sectional dimension of the column. The length l_0 shall not be less than (i) one-sixth of the clear height of the column, (ii) the maximum cross-sectional dimension of the column, and (iii) 450 mm.
- (c) The first tie shall be within a distance equal to $0.5s_0$ from the face of the joint.
- (d) The tie spacing shall not exceed $2s_0$ in any part of the column.

8.3.10 Requirements for Intermediate Moment Frames

8.3.10.1 Scope

For structures assigned to SDC C, structural frames proportioned to resist forces induced by earthquake motions shall satisfy the requirements of Sec 8.3.10 in addition to those of Chapter 6.

8.3.10.2 Reinforcement requirements

Reinforcement details in a frame member shall satisfy 8.3.10.4 below if the factored compressive axial load for the member does not exceed $0.1A_g f'_c$. If the factored compressive axial load is larger, frame reinforcement details shall satisfy Sec 8.3.10.5 below unless the member has spiral reinforcement according to Eq. 6.6.12. If a two-way slab system without beams is treated as part of a frame resisting earthquake effect, reinforcement details in any span resisting moments caused by lateral force shall satisfy Sec 8.3.10.6 below.

8.3.10.3 Shear requirements

Design shear strength of beams and columns resisting earthquake effect, E , shall not be less than the smaller of (i) sum of the shear associated with development of nominal moment strengths of the member at each restrained end of clear span and the shear calculated for factored gravity loads, or (ii) maximum shear obtained from design load combinations that include E , with the E assumed to be twice that prescribed by this Code.

8.3.10.4 Beams

- (a) The positive moment strength at the face of the joint shall not be less than one-third the negative moment strength provided at that face (Figure 6.8.17). Neither the negative nor positive moment strength at any section along the length of the member shall be less than one-fifth of the maximum moment strength provided at the face of either joint.
- (b) At both ends of the member, stirrups shall be provided over lengths equal to twice the member depth measured from the face of the supporting member toward midspan (Figure 6.8.18). The first stirrup shall be located not more than 50 mm from the face of the supporting member. Maximum stirrup spacing shall not exceed (a) $\frac{d}{4}$ (b) 8 times the diameter of the smallest longitudinal bar enclosed, (c) 24 times the diameter of the stirrup bar, and (d) 300 mm.
- (c) Stirrups shall be placed at not more than $\frac{d}{2}$ throughout the length of the member.

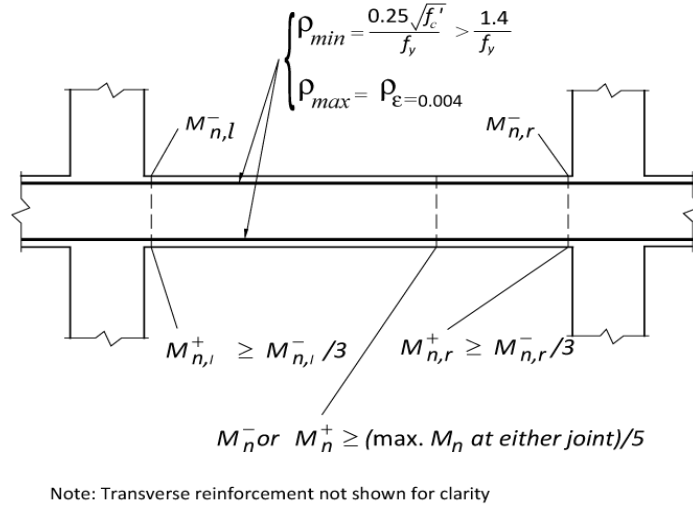


Figure 6.8.17 Flexural requirements for beams (IMF)

8.3.10.5 Columns

- Maximum tie spacing shall not exceed s_0 over a length l_0 measured from the joint face. The spacing s_0 shall not exceed (i) 8 times the diameter of the smallest longitudinal bar enclosed, (ii) 24 times the diameter of the tie bar, (iii) one-half of the smallest cross-sectional dimension of the frame member, and (iv) 300 mm. The length l_0 shall not be less than (i) one-sixth of the clear span of the member, (ii) maximum cross-sectional dimension of the member, and (iii) 450 mm.
- The first tie shall be located not more than $\frac{s_0}{2}$ from the joint face.
- Joint reinforcement shall conform to Sec 6.4.9.
- Tie spacing shall not exceed $2s_0$ throughout the length of the member.

These requirements are shown in Figure 6.8.19.